Algorithms: Simple Guide for the Team

Project: Skin Cancer Detection Using Deep Learning

1. What does our model have to do?

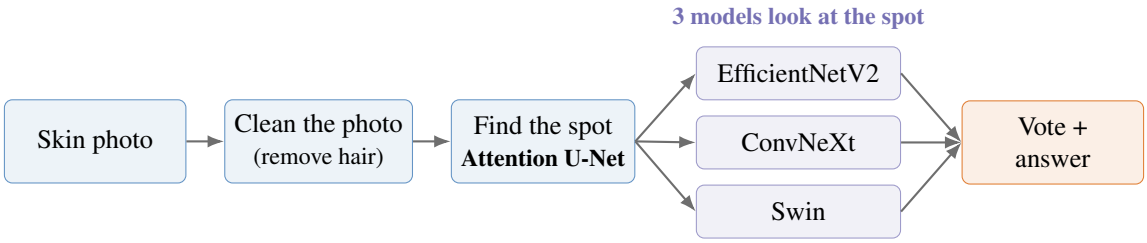
Look at a skin photo and answer two questions:

1. **Where is the lesion?** This is called *segmentation*: drawing an outline around the spot.
2. **What is it?** This is called *classification*: choosing one of 8 labels, e.g., melanoma, BCC, or harmless mole.

2. Words you need to know

Word	Simple meaning
Neural network / model	A program that learns patterns from examples instead of being told the rules
Training	Showing the model thousands of labelled photos so it learns
CNN	A model that scans the photo with small windows to find edges, colours, and textures
Transformer	A model that cuts the photo into small squares and compares every square with every other square to see the big picture
Pre-trained	A model that has already learned general images (dogs, cars, . . .) from ImageNet; we then teach it skin photos. This saves a lot of time and data
Ensemble	Several models vote together, like asking three doctors instead of one
Parameters	The "knobs" inside a model. More knobs = bigger and slower model
Accuracy	Percentage of photos the model gets right
Balanced accuracy	Accuracy averaged over each class, so rare cancers count as much as common moles
Sensitivity	Out of all real cancers, how many did the model catch? (Most important for us)

3. Our pipeline in one picture



The answer contains: the diagnosis, how sure the model is, a **heatmap** (Grad-CAM) showing which part of the photo it looked at, and whether the patient should see a doctor.

4. Algorithms we USE

Attention U-Net (finds the lesion) — WE USE THIS

Simple idea	A U-shaped network: the first half shrinks the photo to understand it, the second half grows it back to draw the outline. "Attention" tells it to ignore healthy skin and hair.
For our project	Step 1. It draws the lesion outline so we can cut out the spot. It outlined lesions better than the plain U-Net in a fair test.
Paper	https://arxiv.org/abs/1804.03999
Code	https://github.com/qubvel-org/segmentation_models.pytorch

EfficientNetV2 (model 1 — a CNN) — WE USE THIS

Simple idea	A CNN designed to be accurate *and* fast by growing its depth, width, and image size in a balanced way.
For our project	EfficientNet models won two big skin-cancer competitions (ISIC 2019 and 2020). Medium size (about 22M parameters).
Paper	https://arxiv.org/abs/2104.00298
Code	https://github.com/huggingface/pytorch-image-models

ConvNeXt (model 2 — a modern CNN) — WE USE THIS

Simple idea	An older CNN (ResNet) redesigned with tricks borrowed from transformers. It thinks like a transformer but runs like a CNN.
For our project	A group of ConvNeXt models gave the best reported result on the 8-class ISIC 2019 data.
Paper	https://arxiv.org/abs/2201.03545
Code	https://github.com/facebookresearch/ConvNeXt

Swin Transformer (model 3 — a transformer) — WE USE THIS

Simple idea	A transformer that looks at the photo in small windows and then shifts the windows, so it sees both small details and the whole shape.
For our project	CNNs are good at texture; Swin is good at overall shape and symmetry. Because they make *different* mistakes, voting together makes the team stronger.
Paper	https://arxiv.org/abs/2103.14030
Code	https://github.com/microsoft/Swin-Transformer

Focal loss (training trick) — WE USE THIS

Simple idea	A way of scoring mistakes that punishes the model more when it gets *rare* or *hard* photos wrong.
For our project	Cancer photos are rare. Without this, the model would mostly learn "harmless mole". In one study it added about 9 points of accuracy.
Paper	https://arxiv.org/abs/1708.02002
Code	A few lines of PyTorch (torchvision has a 2-class version: `sigmoid_focal_loss`)

Grad-CAM (explanation heatmap) — WE USE THIS

Simple idea	Colours the photo red where the model paid most attention.
For our project	Lets the doctor check that the model looked at the lesion, not at a ruler or hair. Builds trust.
Paper	https://arxiv.org/abs/1610.02391
Code	https://github.com/jacobgil/pytorch-grad-cam

5. Algorithms we compare against (baselines)

We also train these simpler, well-known models so we can *prove* our method is better.

ResNet-50 — BASELINE

Simple idea	A CNN with "shortcut" connections that let it be very deep without getting confused. The most common starting model.
For our project	Standard baseline. 25.6M parameters.
Paper	https://arxiv.org/abs/1512.03385
Code	https://github.com/huggingface/pytorch-image-models

DenseNet-121 — BASELINE

Simple idea	Every layer passes its findings to all later layers, so information is reused. Small (8M parameters).
For our project	Baseline; popular in skin-cancer papers.
Paper	https://arxiv.org/abs/1608.06993
Code	https://github.com/huggingface/pytorch-image-models

MobileNetV2 — BASELINE

Simple idea	A tiny CNN built to run on mobile phones (3.5M parameters).
For our project	Baseline, and a candidate for a future phone app.
Paper	https://arxiv.org/abs/1801.04381
Code	https://github.com/huggingface/pytorch-image-models

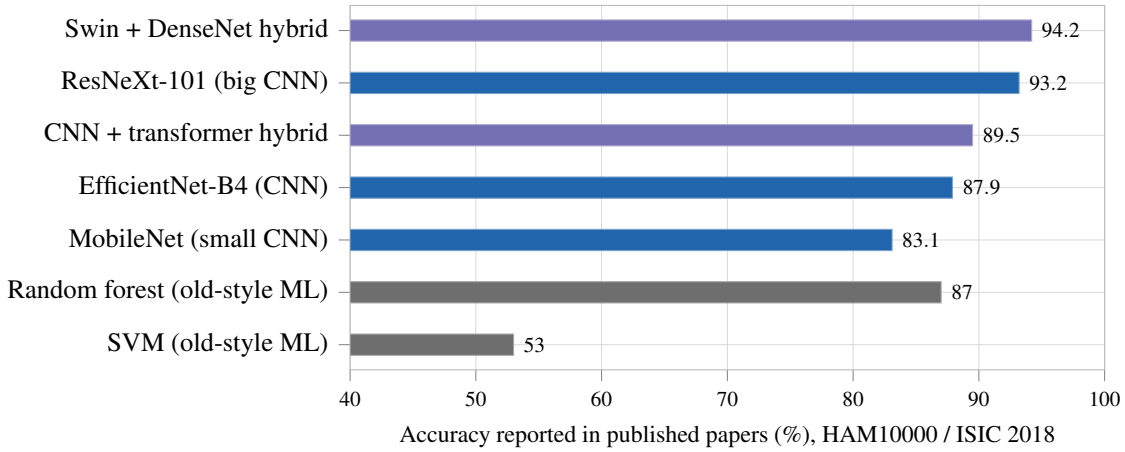
DINOv2 (quick strong baseline) — BASELINE

Simple idea	A big model that learned from 142 million unlabelled images. We only add a small final layer on top, so it is cheap to try.
For our project	Quick check of how far a ready-made model can go.
Paper	https://arxiv.org/abs/2304.07193
Code	https://github.com/facebookresearch/dinov2

6. Other algorithms (good to know)

Algorithm	Simple idea / why not our main choice	Paper
SVM / Random Forest	Old-style machine learning with hand-made features. Much weaker than deep learning	–
VGG16	Early simple CNN. Very big (138M parameters) for its accuracy	arxiv.org/abs/1409.1556
InceptionV3	CNN with filters of several sizes; used in the famous 2017 Nature skin-cancer paper	arxiv.org/abs/1512.00567
Xception	Faster version of Inception	arxiv.org/abs/1610.02357
EfficientNet (V1)	Older version of EfficientNetV2	arxiv.org/abs/1905.11946
ViT	The first image transformer. Needs a lot of data, so Swin is better for us	arxiv.org/abs/2010.11929
U-Net	The original outline model (Attention U-Net is its upgrade)	arxiv.org/abs/1505.04597
TransUNet	U-Net with a transformer. Scored lower than Attention U-Net in a fair test	arxiv.org/abs/2102.04306
UCTransNet	Another U-Net upgrade; slightly better but more complex	arxiv.org/abs/2109.04335
PanDerm	A huge skin “foundation model” (2025). Very strong but needs big computers	github.com/SiyuanYan1/PanDerm

7. How the algorithms compare (one simple chart)



What the chart tells us: deep learning is far better than old-style machine learning, and mixing CNNs with transformers gives the best results. **Careful:** these numbers come from different papers with slightly different setups, so treat them as a rough guide. Real-world results are usually lower. That is why we test on outside datasets.

8. Tools we will install

Tool	What for	Link
PyTorch	The main deep-learning library	pytorch.org
timm	Ready-made EfficientNetV2, ConvNeXt, Swin, ResNet, DenseNet, MobileNet	github.com/huggingface/pytorch-image-models
segmentation-models-pytorch	Ready-made U-Net for finding the lesion	github.com/qubvel-org/segmentation_models.pytorch
pytorch-grad-cam	Heatmap explanations	github.com/jacobgil/pytorch-grad-cam
scikit-learn	Data splitting and scores	scikit-learn.org

Summary: Attention U-Net finds the spot → EfficientNetV2 + ConvNeXt + Swin each look at it → they vote → we show the answer with a Grad-CAM heatmap. Focal loss helps with rare cancers. ResNet-50, DenseNet-121, MobileNetV2, and DINOv2 are our baselines. The detailed version is in `Algorithms_Report_Detailed.pdf`.